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Oefeningenreeks 5D nr. 13

$$y^2 = x^2 - x^4$$

$$\Rightarrow 4 \int_0^1 \sqrt{x^2 - x^4} dx = 4 \int_0^1 x \sqrt{1 - x^2} dx$$

$$u = 1 - x^2$$

$$du = -2x dx$$

$$= 4 \int_0^1 \frac{-1}{2} \sqrt{u} du = 2 \int_0^1 x \sqrt{u} du$$

$$= 2 \left[\frac{2u^{3/2}}{3} \right]_0^1$$

$$= \frac{4 \cdot 1}{3} - \frac{4 \cdot 0}{3}$$

$$= \frac{4}{3}$$

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References